

Homework 7

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November 26, 2025

Exercise 1

We test the counterfactual probability $\theta' = \frac{1}{3}$ against the true probability of $\theta = \frac{1}{2}$. The learning algorithm learns the true probability when the observed frequency is within $\frac{|1/2 - 1/3|}{2} = \frac{1}{12}$ of θ . Under the IID assumption, with $n = 150$ samples, this occurs with probability at least:

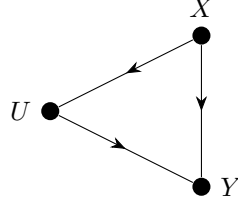
$$P_{1/2,150}(|\text{observed frequency} - 1/2| \leq \frac{1}{12}) \geq 1 - \frac{1}{4(150)(1/12)^2} = 0.76$$

Similarly, with $n = 500$ samples:

$$P_{1/2,500}(|\text{observed frequency} - 1/2| \leq \frac{1}{12}) \geq 1 - \frac{1}{4(500)(1/12)^2} = 0.928$$

Exercise 2

Consider the Bayesian network with the following DAG over Boolean variables:



with the following probability tables:

$P(X = 0)$	$P(X = 1)$
$1/2$	$1/2$

x	$P(U = 0 X = x)$	$P(U = 1 X = x)$
0	$9/10$	$1/10$
1	$1/5$	$4/5$

u	x	$P(Y = 0 U = u, X = x)$	$P(Y = 1 U = u, X = x)$
0	0	1	0
0	1	1	0
1	0	$1 - q$	q
1	1	$1 - (1 - r)q$	$(1 - r)q$

We first compute $P(Y = y|X = x)$.

$$\begin{aligned}
 P(Y = y|X = x) &= P(Y = y, U = 0|X = x) + P(Y = y, U = 1|X = x) \\
 &= P(Y = y|U = 0, X = x)P(U = 0, X = x) \\
 &\quad + P(Y = y|U = 1, X = x)P(U = 1, X = x)
 \end{aligned}$$

x	$P(Y = 0 X = x)$	$P(Y = 1 X = x)$
0	$1 - q/10$	$q/10$
1	$1 - 4(1 - r)q/5$	$4(1 - r)q/5$

Similarly, we compute $P(Y)$.

$$\begin{aligned}
 P(Y = y) &= P(Y = y, X = 0) + P(Y = y, X = 1) \\
 &= P(Y = y|X = 0)P(X = 0) + P(Y = y|X = 1)P(X = 1)
 \end{aligned}$$

$P(X = 0)$	$P(X = 1)$
$1 - q/20 - 2(1 - r)q/5$	$q/20 + 2(1 - r)q/5$

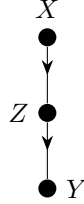
Setting $P(Y = 1|X = 0)$ and $P(Y = 1)$ equal:

$$\begin{aligned}q/10 &= q/20 + 2(1-r)q/5 \\q/20 - 2(1-r)q/5 &= 0 \\q(1/20 - 2(1-r)/5) &= 0 \\q(r - 7/8) &= 0\end{aligned}$$

Either $q = 0$ (and r takes on any value) or $r = 7/8$ (and q takes on any value).

Exercise 3

Consider the Bayesian network N_1 with the following DAG over Boolean variables:



with the following probability tables:

$P(X = 0)$	$P(X = 1)$
1/2	1/2

x	$P(Z = 0 X = x)$	$P(Z = 1 X = x)$
0	1/3	2/3
1	2/3	1/3

z	$P(Y = 0 Z = z)$	$P(Y = 1 Z = z)$
0	1/6	5/6
1	5/6	1/6

We then compute $P(Y = y, Z = z, X = x)$.

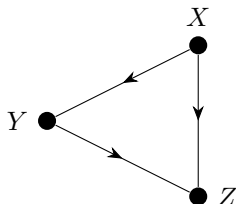
$$\begin{aligned}
 P(Y = y, Z = z, X = x) &= P(Y = y|Z = z, X = x)P(Z = z, X = x) \\
 &= P(Y = y|Z = z)P(Z = z, X = x) \\
 &= P(Y = y|Z = z)P(Z = z|X = x)P(X = x)
 \end{aligned}$$

Constructing the joint probability distribution:

y	z	x	$P(Y = y, Z = z, X = x)$
0	0	0	$1/6 \cdot 1/3 \cdot 1/2 = 1/36$
0	0	1	$1/6 \cdot 2/3 \cdot 1/2 = 1/18$
0	1	0	$5/6 \cdot 2/3 \cdot 1/2 = 5/18$
0	1	1	$5/6 \cdot 1/3 \cdot 1/2 = 5/36$
1	0	0	$5/6 \cdot 1/3 \cdot 1/2 = 5/36$
1	0	1	$5/6 \cdot 2/3 \cdot 1/2 = 5/18$
1	1	0	$1/6 \cdot 2/3 \cdot 1/2 = 1/18$
1	1	1	$1/6 \cdot 2/3 \cdot 1/2 = 1/36$

Exercise 4

Consider the Bayesian network N_2 with the following DAG over Boolean variables:



Note that:

$$P(Y = y|X = x) = \frac{P(Y = y, X = x)}{P(X = x)}$$

$$P(Z = z|Y = y, X = x) = \frac{P(Z = z, Y = y, X = x)}{P(Y = y, X = x)}$$

We can then construct the following probability tables:

$P(X = 0)$	$P(X = 1)$
1/2	1/2

x	$P(Y = 0 X = x)$	$P(Y = 1 X = x)$
0	11/18	7/18
1	7/18	11/18

y	x	$P(Z = 0 Y = y, X = x)$	$P(Z = 1 Y = y, X = x)$
0	0	1/11	10/11
0	1	2/7	5/7
1	0	5/7	2/7
1	1	10/11	1/11

We then compute $P(Z = z, Y = y, X = x)$:

$$P(Z = z, Y = y, X = x) = P(Z = z|Y = y, X = x)P(Y = y, X = x)$$

$$= P(Z = z|Y = y, X = x)P(Y = y|X = x)P(X = x)$$

Constructing the joint probability distribution:

z	y	x	$P(Z = z, Y = y, X = x)$
0	0	0	$1/11 \cdot 11/18 \cdot 1/2 = 1/36$
0	0	1	$2/7 \cdot 7/18 \cdot 1/2 = 1/18$
0	1	0	$5/7 \cdot 7/18 \cdot 1/2 = 5/36$
0	1	1	$10/11 \cdot 11/18 \cdot 1/2 = 5/18$
1	0	0	$10/11 \cdot 11/18 \cdot 1/2 = 5/18$
1	0	1	$5/7 \cdot 7/18 \cdot 1/2 = 5/36$
1	1	0	$2/7 \cdot 7/18 \cdot 1/2 = 1/18$
1	1	1	$1/11 \cdot 11/18 \cdot 1/2 = 1/36$

Exercise 5

Note that X and Y are independent in N_1 . Since N_1 and N_2 have the same joint probability distribution table, this also holds in N_2 . However, there is an edge between X and Y in N_2 , so this does not hold for all Bayesian networks with the same DAG. Consequently, N_2 is not faithful.